JOINT UNIVERSITIES PRELIMINARY EXAMINATIONS BOARD

AUGUST 2022 EXAMINATIONS

OPTION – D301

PHYSICS

Time Allowed: 3 hours

SECTION A: MULTIPLE CHOICE QUESTIONS

Answer ALL questions in this section.

Use the OMR answer sheet provided to answer the questions.

Write and shade on your OMR Sheet the OPTION written on your question paper

Follow the instructions on the OMR sheet.

SECTION B: ESSAY QUESTIONS

Answer FOUR Questions in ALL; ONE Question from each course.

Ensure you read and follow all the Instructions on the cover page of the Answer Booklet.

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3. The velocity of a moving train is given by $V = \frac{P}{t} + Qt$ where V is the velocity and t is the time. Find the SI unit of P and Q respectively.

A. ms^{-1}, ms^{-3}.

B. ms^{-1}, ms^{-1}.

C. m, ms^{-2}.

D. m, ms^{2}.

4. A train goes into the tunnel at 20m/s and emerges from it at 55m/s. The tunnel is 1500m long, assuming constant acceleration. How long did the train stay in the tunnel?

A. 75s.

B. 40s.

C. 27.3s.

D. 20s.

5. The main source of energy used for satellites is

A. Edison cells.

B. Fuel cells.

C. Solar cells.

D. Cryogenic cells.

6. The gravitational force between two objects of masses 10^{24} kg and10^{27} kg is 6.6 N. Calculate the distance between them.

A. 1.0×10^{19} m.

B. 1.0×10^{-19}m.

C. 1.0×10^{20}m.

D. 1.0×10^{-20}m.

7. The magnetic flux in a coil having 500 turns changes at the time rate of 0.04 Wbs^{-1}. What is the induced e.m.f in the coil?

A. 50 V.

B. 35 V.

C. 20 V.

D. 10 V.

13. Calculate the de Broglie wavelength of an electron with energy 1.0 keV.

A. 3.9×10^{-11} m.

B. 4.5×10^{-11} m.

C. 7.4×10^{-11} m.

D. 9.2×10^{-11} m. $[1eV = 1.6 \times 10^{-19}J]$

14. Which of the following has the most penetrating power?

A. alpha-particles.

B. beta-particles.

C. gamma-rays.

D. X-rays.

15. The stopping potential V_s was plotted against the frequency of radiation f on a Cartesian plane. Determine the intercept.

A. $\frac{h}{e}$.

B. $\frac{w_o}{e}$.

C. $\frac{e}{m}$.

D. E.

16. A rocket of mass 100 kg was lunched from the surface of the earth so that it escaped the gravitational attraction of the earth. Calculate its escape velocity.

A. 1.12×10^4 m/s.

B. 1.11×10^4 m/s.

C. 6.00×10^{27} m/s.

D. 3.84×10^{34} m/s. $[g = 9.8\ m/s^2; R_E = 6.4 \times 10^6\ m;$ and $M_E = 6 \times 10^{24}\ kg]$

17. A body of mass 10 kg is acted on by a constant force of 8 N for 5 seconds. Calculate the kinetic energy gained by the body at the end of the time.

A. 40.0J.

B. 60.0J.

C. 70.0J.

D. 80.0J.

23. A given mass of gas at 40 °C is trapped in a tube. If its volume is reduced by one-third of its initial value by applying a pressure which surpasses its initial pressure by 60%, calculate the new temperature of the gas.

A. 125.2 °C.

B. 60.9 °C.

C. 62.6 °C.

D. 333.9 °C.

24. The fundamental frequency of vibration of a stretched wire is 120Hz. What will be the new fundamental frequency if the tension in the wire is doubled with the length remaining constant?

A. 240Hz.

B. 60Hz.

C. 140Hz.

D. 170Hz.

25. According to Huygens' principle, the position of light wave at any point in time is given by the

A. wavelength.

B. waveguide.

C. wavefront.

D. wave number.

26. Doppler effect can be described as the

A. superposition of sound wave due to relative motion of the source and the observer.

B. change in wavelength due to reflective motion of the source and the observer.

C. change in pitch produced by the relative motion of the source and the observer.

D. change in amplitude produced by the relative motion of the source and the observer.

27. A small spherical ball bearing of radius 5cm is placed in a uniform electric field of 150 N/C. Calculate the maximum electric flux that can pass through it.

A. 4.7 Nm^2/C.

B. 3.0 Nm^2/C.

C. 1.5 Nm^2/C.

D. 0.1 Nm^2/C.

33. The instrument used in marking the upper fixed point of thermometer is known as
    A. thermometer.
    B. manometer.
    C. spectrometer.
    D. hypsometer.

34. A pair of eyeglasses frame is made of epoxy plastic. At room temperature (20.0 °C), the frame has circular lens holes 2.20 cm in radius. To what temperature must the frame be heated if lenses 2.21 cm in radius are to be inserted in them? The average coefficient of linear expansion for epoxy is $1.30 \times 10^{-4}\ °C^{-1}$.
    A. 58 °C.
    B. 56 °C.
    C. 55 °C.
    D. 57 °C.

35. Latent heat of vaporization is the amount of heat required to
    A. change phase of a substance from solid to liquid without any change in temperature.
    B. change phase of a substance from liquid to gas without any change in temperature.
    C. raise temperature of a 1 kg of a substance by 1 K.
    D. raise temperature of a substance by 1 K.

36. The temperature of a wire of length 10m is 15°C. If the wire is heated to 65°C, calculate the new length of the wire correct to 3 decimal places.
    A. 10.023 m.
    B. 10.012 m.
    C. 10.050 m.
    D. 10.025 m.

    [Linear expansivity of the wire is $1.0 \times 10^{-4}\ K^{-1}$]

37. A Carnot engine operates between temperature limits of 473K and 293K. What is its maximum possible efficiency?
    A. 38 %
    B. 72 %
    C. 90 %
    D. 100%

43. An electric iron is rated 1 kW, 230 V. What is the resistance of its element?

A. 18.9 Ω.

B. 4.3 Ω.

C. 52.9 Ω.

D. 55.9 Ω.

44. Which of the following does not correctly defines dielectric strength? It is the

A. maximum electric field intensity that a pure material can withstand under ideal conditions without breaking down.

B. measure of the electrical strength of an insulator.

C. maximum voltage that can be applied to a given material without causing it to break down.

D. ratio of the permittivity of free space to the permittivity of the dielectric material.

45. A lithium nucleus ${}^{7}_{3}Li$ travelling at 2.0×10^{5} m/s enters a region of uniform magnetic field of flux density 0.10 T. What is the maximum magnetic force experienced by this nucleus?

A. 3.2×10^{-15}N.

B. 2.2×10^{-14} N.

C. 3.2×10^{-14}N.

D. 9.6×10^{-15}N.

46. Two straight parallel conductors carry equal currents in the same direction. The force between them is

A. zero.

B. equal.

C. repulsive.

D. attractive.

47. What resistance is required to convert a galvanometer of internal resistance 10 Ω and full scale deflection of 20 mA to a voltmeter that can measure up to 15 V?

A. 750 Ω.

B. 740 Ω.

C. 200 Ω.

D. 75 Ω.

SECTION B: ESSAY QUESTIONS

Answer FOUR Questions in ALL; ONE Question from each Course.

PHY 001: MECHANICS AND PROPERTIES OF MATTER

1. (a) (i) Define the following terms:

Young modulus, Bulk modulus; and Shear modulus. **[3 Marks]**

(ii) Two parallel oppositely directed forces, each 3500 N, are applied tangentially to the upper and lower faces of a cubical metal block 25 cm on a side. Find the angle of shear and displacement of the upper surface relative to the lower surface if the shear modulus for the metal is 80 GPa.

[3 Marks]

(b) An airplane is travelling horizontally at a speed of 60 m/s and drops a crate of emergency supplies. To avoid damage, the maximum vertical speed of the crate on landing is 30 m/s. Assuming air resistance is negligible,

(i) calculate the maximum height of the airplane when the crate is dropped;

(ii) calculate the time taken for the crate to reach the ground from this height.

(iii) If the airplane is travelling at the maximum permitted height, calculate the horizontal distance travelled by the crate after it is released from the airplane.

[4 Marks]

2. (a) Define the following:

(i) terminal velocity, (ii) viscosity, (iii) surface tension. **[3 Marks]**

(b) State the following:

(i) Stokes' law, (ii) law of conservation of angular momentum. **[3 Marks]**

(c) A mass of 2 kg attached to the end of a vertical wire of length 2 m and diameter 2 mm extended the wire by 1 mm. Calculate the

(i) Young's modulus of the wire,

(ii) energy stored in the wire.

[4 Marks]

5. (a) Explain the following terms:

(i) Electric flux (ii) Equipotential surface. **[2 Marks]**

(b) (i) State the condition for resonance in RLC circuits. **[1 Mark]**

(ii) A series RLC circuit with $R = 425\Omega$, $L = 1.25H$, and $C = 3.50\mu F$, is connected to a maximum voltage supply, $V = 150V$, with $\omega = 377\ s^{-1}$. Determine the impedance and maximum current in the circuit.

[5 Marks]

(c) The maximum power dissipated in a $5k\Omega$ resistor is 15W. What is the maximum current?

[2 Marks]

6. (a) Explain the statement: "The capacitance of a capacitor is 5 farads." **[2 Marks]**

(b) A capacitor consisting of 2 parallel plates separated by oil of dielectric constant 1.5, each plate having an effective area of $1000cm^2$ and spaced 0.1cm apart, is connected across a constant voltage source of 5000V. Calculate:

(i) the capacitance of the capacitor, **[1 Mark]**

(ii) the charge of the capacitor, **[1 Mark]**

(iii) the energy stored in the capacitor, and **[1 Mark]**

(iv) If the voltage source was doubled, while the capacitance was kept constant, what would be the new amount of energy stored?

[2 Marks]

(c) Two charges $Q_1 = +200\ \mu C$ and $Q_2 = -100\mu C$ are separated by a distance of 100cm, with Q_1 located at the origin. How much work must be done to transfer a charge $Q_3 = +500\mu C$ from a point x = 80cm to a point x = 20cm?

[3 Marks]